Water Quality Exam

MIT Science Olympiad Invitational

Jan. 25, 2020

Team Number: _____

Team Name: _____

Participant 1: _____

Participant 2: _____

____/300



hi welcome to water quality :)

Materials:

- one sheet of notes, two non-programmable, non-graphing calculators, two pairs of goggles, salinometer, pencils, erasers, & your brain!!

Instructions:

- You may **write** on the exam, but only the answer sheet will be **graded**. Please **DO NOT** write on the image sheet.
- You may **split** the exam, answer sheet, and image sheet apart, but be sure to write your team name and number on every sheet of the exam and answer sheet.
- You will be called up for **Part IV: Salinometer Testing** by team number. You will get **4 minutes** to test both solutions.
- **a few notes:** test is a lil bit long so be sure to prioritize & skip sections !! questions are not ordered by level of difficulty; tiebreaker order is on answer sheet for reference.
- good luck good luck :))

questions? comments?
email me at emfang@mit.edu :)

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Part I: Marine and Estuary Ecology (30%)

I. Multiple Choice (2 pts each, 30 pts total)

1. **Which of the following is not a method of preventing sediment pollution?**
    a. Allowing soil to be rich in hummus
    b. Allowing several different types of plants to grow together
    c. Allowing indigenous plants to grow along riverbanks
    d. Allowing cattle to overgraze, and then switching fields

2. **Which area has the most run-off after a storm?**
    a. Urban
    b. Agricultural
    c. Suburban
    d. Forest

3. **Aquifers are directly supplied water by which of the following processes?**
    a. Infiltration
    b. Runoff
    c. Transpiration
    d. Condensation

4. **What type of intermolecular forces increase surface tension in water?**
    a. London dispersion
    b. Hydrogen bonding
    c. Dipole-dipole
    d. Ion-dipole

5. **How many hydrogen bonds can a molecule of water participate in?**
    a. 1
    b. 2
    c. 3
    d. 4

6. **What percentage of energy in an organism is passed onto the next trophic level?**
    a. 5%
    b. 10%
    c. 20%
    d. 50%

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7. **Which of the following is an example of predator partitioning in a Chesapeake Bay ecosystem?**
    a. Bald eagles and ospreys both consume large piscivorous fish.
    b. If the bald eagle disappeared, ospreys would have greater access to piscivorous fish.
    c. Snakes only consume rats, while cranes only consume frogs.
    d. Frogs are more likely to survive to reproduce while rats are less likely to survive to reproduce when snakes are present.

8. **Refer to Diagram 1 on your image sheet. In which of the following regions of a coral reef stabilizes the reef from the force of large waves?**
    a. A
    b. B
    c. C
    d. D
    e. E
    f. F
    g. G

9. **Which of the following best describes the bottleneck effect?**
    a. A few individuals in the population colonize a new, isolated habitat
    b. A catastrophe kills off most of the population at random and leaves few survivors
    c. The population exceeds carrying capacity and death rates begin to surpass birth rates
    d. The population jumps from one steady state to another, with no intermediates

10. **Some algae live on the shells of turtles. This is an example of:**
    a. Obligate mutualism
    b. Facultative mutualism
    c. Commensalism
    d. Ammensalism

11. **A lion catches the last remaining gazelle in the grassland, causing other lions to go without food for the day. What type of competition is this?**
    a. Apparent
    b. Interference
    c. Intraspecific
    d. Interspecific

12. **The Haber-Bosch process fixes nitrogen into what molecule?**
    a. NO_2^-
    b. NO_3^-
    c. NH_3
    d. NH_4^+

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13. In which of the following places can phosphorus be found naturally?

a. Nucleic acids
b. Phospholipids
c. Rocks
d. All of the above

For questions 9-10, refer to Diagram 2 on your Image Sheet:

14. What type of estuary is depicted in Image (b)?

a. Bar-built
b. Coastal-plain
c. Fjord
d. Tectonic

15. The estuaries in the above image are classified by what features? What other features can estuaries be classified by?

a. Bathymetry, topography
b. Topography, bathymetry
c. Water circulation, geology
d. Geology, water circulation

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II. Scenario: The Chesapeake Bay (30 pts)

1. What type of estuary is the Chesapeake Bay? (1)
2. How was the Chesapeake Bay estuary formed? (1)

Many tributaries in Virginia flow into the Chesapeake Bay. Diagram A shows acres of submerged vegetation in Virginia over time.

3. What is the trend in submerged vegetation from 2010-2012? (1)
4. What is the effect of this trend on… (2)
    a. Turbidity (1)
    b. Dissolved oxygen (1)
5. On your answer sheet is a life table for a sample population of the Chesapeake Bay blue crab during this time. Answer the following questions. The following key may be helpful to you:

x	Stage/class
a_x	Number of clams observed at each stage/class
l_x	Survivorship
d_x	Mortality
q_x	Mortality rate for each stage/class
F_x	Total fecundity for each stage/class
m_x	Individual fecundity
$l_x m_x$	Number of offspring produced per original individual during each stage or class;
r_o	Basic reproductive rate

    a. Fill in the empty cells on the life table on your answer sheet. Show work (including equations and substitutions) for the **cell in each column that is shaded gray** in the boxes the table**.** Round all answers to **two decimal places.** Each column is worth 3 points—1 point for correctly showing work, 1 point for a correct answer in the shaded box, and 1 points for correct answers in all other boxes*.* Calculating **r_o** is worth 2 points. 17 pts total.
    b. Is the crab population increasing or decreasing? Use information from the life table to support your answer (2).
    c. What is one possible anthropological cause for this increase or decrease in population? (1)

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Diagram B shows a typical food web seen in the Chesapeake Bay.

6. What effect will the increase or decrease of the blue crab population (from #5b) have on the population of… (2)
    a. Zooplankton (1)
    b. Herring menhaden (1)
7. Another issue in the Chesapeake Bay is the overfishing of striped bass. What effect will this have on the population of…(3)
    a. Raptors (1)
    b. Herring menhaden (1)
    c. Clams (1)

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III. Scenario: Local water pollution (30)

You identify a group of farms that have been excessively using fertilizer and a local factory that has been dumping sewage in Virginia, near the James River, a Chesapeake Bay tributary.

1. Is the factory considered a point source or a non-point source of pollution? (1)
2. Is run-off from the farm fertilizers considered a point source or a non-point source of pollution? (1)

You decide that the wastewater from the factory needs to be treated. Diagram C shows the process of wastewater treatment.

3. In the table on your answer sheet, label the missing steps of wastewater treatment in Diagram C and states the purpose of step/process. (12)

In addition, you decide that you want to make the water in the river potable.

4. Naturally, you begin with the first two steps of potable water treatment: coagulation and flocculation. Briefly describe the purposes and processes of coagulation and flocculation, as well as factors you would consider in selecting a coagulant and a mixing speed. *(10 pts)*
5. You encounter an issue during sedimentation: the particles are not settling. List and explain three reasons why this could be happening. *(6 pts)*

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Part II: Coral Reef Macroflora and Fauna Identification (30%)

I. Organism Identification & Short Answer (60)

Answer each of the following questions (and subquestions) using the first page of the Image Sheet. Point values are placed next to each question.

1. **(4 pts total)**
    a. What organism is depicted in Image O? (1)
    b. What is another common name for this organism? (1)
    c. In what region is this organism found? (1)
    d. How does this organism feed? (1)
2. **(4 pts total)**
    a. What organism is depicted in Image I? (1)
    b. What is the diet of this organism? (1)
    c. Name one adaptation that these organisms have to help themselves eat (1).
    d. Name one other organism on this year's list of Macroflora and Fauna list that has the same diet (1).
3. **(5 pts total)**
    a. What organism is depicted in Image J? (1)
    b. What types of waters do these organisms live in? (2)
    c. What do these organisms eat? (1)
    d. How do the swimming patterns of these organisms help them survive? (1)
4. **(7 pts total)**
    a. Which image depicts a long-spined sea urchin? (1)
    b. What is the impact of long-spined sea urchins on coral reef ecosystems (3)?
    c. What are three factors that impede the repopulation of these sea urchins in lowly-populated coral reefs? (3)
5. **(7 pts total)**
    a. Which image depicts a Nassau grouper? (1)
    b. What habitat does this organism occupy? (1)
    c. What other organism on this year's list of Macroflora and Fauna is commonly consumed by this organism? (1)
    d. What is the conservation status of this organism? (1)
    e. What reproduction habit of this organism contributes to its conservation status? (3)
6. **(6 pts total)**
    a. Which image depicts a moray eel? (1)
    b. What is the common name for the specific species depicted in the image? (1)
    c. What anatomical feature makes this organism unique from all other animals? Explain the function of this feature. (3)
    d. Why does consuming this organism lead to food poisoning? (1)

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7. **(7 pts total)**
    a. What organism is depicted in Image K? (1)
    b. What type of relationship does this organism have with many types of fish? (1)
    c. Describe how both organisms (the organism you identified, and a fish) benefit from/are hurt by this relationship (3).
    d. How do the organisms (organism you identified and a fish) communicate with each other? (2)
8. **(6 pts total)**
    a. Which image depicts a humphead wrasse? (1)
    b. Why is this species endangered? (2)
    c. The humphead wrasse is considered an umbrella species. What does this mean, and what implications does this have on the species around it? (2)
    d. How does the humphead wrasse reproduce? (1)
9. **(8 pts total)**
    a. Which image depicts a flamingo tongue snail? (1)
    b. How do these organisms' feeding habits defend themselves against predators? (2)
    c. In addition, this organism also uses its bright colors to warn its predators. What is this process called? (1)
    d. However, the bright colors of this organism have proven to be detrimental to their conservation status. How so? (2)
    e. What is the relationship between this organism and coral reefs? (2)
10. Six of the images on the Image Sheet were not used or referenced in any of the above questions. List those images on the answer sheet, and identify the organism depicted in those images (6).

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II. Scenario: The Great Barrier Reef (30 pts)

Diagram D shows mass coral bleaching on the Great Barrier Reef in 2016 and 2017. Prior to 2017, mass bleaching had only occurred four times in the history of the Great Barrier Reef.

1. In your own words, describe what coral bleaching is (3).
2. What is the significance of the two mass bleachings occuring in back-to-back years? (1)
3. The mass bleaching in 2017 occurred even in the absence of an El Niño event. What is an El Niño event, and why is this significant? (2)

Diagram E shows the number of crown-of-thorns starfish outbreaks since 1998.

4. What is the trend in the number of crown-of-thorns starfish outbreaks since 2009? (1)
5. Listed below are potential causes for crown-of-thorns outbreaks. Please state if each is a valid cause, and explain why/why not it would cause an outbreak *(8 pts total).*
    a. Increased runoff from nearby farms (2)
    b. Overfishing of tritons (2)
    c. Decreased water temperature (2)
    d. Increased phytoplankton population (2)
6. What is the impact of crown-of-thorns starfish on coral reef health? (1)
7. Why is it difficult to control the starfish population? What is currently being done to control the population, and is it effective? Why or why not? (4)

Diagram F shows the location of sugarcane farms, with respect to the Great Barrier Reef. Sugarcane farms are known for using a large amount of fertilizers.

8. What is the significance of the locations of sugarcane farms, with respect to the Great Barrier Reef? (2)
9. What is the impact of fertilizers on water quality, in general? (1)
10. In addition to fertilizer, sugarcane farms also contain a large amount of loose sediment.
    a. How does increased sediment pollution impact water temperature? (1)
    b. What are two effects that sediment pollution has on fish? (2)
11. How do the presence of sugarcane farms impact the coral-of-thorns starfish population AND coral reef health? Explain. (4)

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Part III: Water Monitoring and Analysis

I. Multiple Choice (2 pts each, 30 pts total)

1. **Suppose you have a sample of water. As total dissolved solids of the sample increases, what happens to the temperature and turbidity of the sample, respectively?**
    a. Increase, increase
    b. Decrease, decrease
    c. Increase decrease,
    d. No change, increase

2. **Above what concentration of E.coli is a water sample considered unsafe to drink?**
    a. 0 colonies/100 mL
    b. 50 colonies/100 mL
    c. 100 colonies/100 mL
    d. 200 colonies/100 mL

3. **As temperature increases, what happens to dissolved oxygen in a water sample?**
    a. Increases
    b. Decreases
    c. No change
    d. Can't predict dissolved oxygen levels from temperature changes

4. **The Winkler Method uses what chemical practice to determine what quality of water?**
    a. Redox reaction, biological oxygen demand
    b. Redox reaction, dissolved oxygen
    c. Titration, biological oxygen demand
    d. Titration, dissolved oxygen

5. **Which of the following is not a cause of turbidity variation in a lake?**
    a. Natural color of the water changes
    b. Nearby mining contributes colloids to the sample
    c. pH of the sample increases due to dissolved OH^-
    d. Seasonal variations cause nutrients to be released from bottom

6. **Which of the following is not an effect of increasing the turbidity of water on organisms?**
    a. Reduction of efficiency of fish gills and respiration
    b. Fewer nutrients and particulate matter for consumption
    c. Decreased primary productivity and photosynthesis
    d. Increased temperature

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7. **Which of the following processes decreases the pH of water?**
    a. Fossil fuel emissions
    b. Volcano emissions
    c. Dissolving limestone
    d. Both A and B

8. **Which of the following will occur when the salinity of a water sample increases?**
    a. Density decreases
    b. Heat capacity decreases
    c. Conductivity decreases
    d. None of the above

9. **Which of the following is not a source of nitrates in water?**
    a. Acid rain
    b. Sewage
    c. Fertilizer
    d. Erosion

10. **Which of the following situations will not increase dissolved oxygen content in a river?**
    a. High altitude
    b. Dam removal
    c. More riparian vegetation
    d. All of the above

11. **Which of the following characteristics are true of nitrates?**
    a. Insoluble in water
    b. Blue in color
    c. Produce a rotten egg smell
    d. None of the above

12. **All of the following are sources for fecal coliform except:**
    a. Wastewater discharge
    b. Livestock runoff
    c. Organic plant debris
    d. Septic tank overflow

13. **Which form of phosphorus is readily available for biological processes without further breakdown in aquatic organisms?**
    a. Phosphates
    b. Orthophosphates
    c. Polyphosphates
    d. Biphosphorus

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14. What unit is turbidity usually measured in?

a. NTU
b. mg/L
c. M
d. ft

15. If most people in a town are suffering from an upset stomach, diarrhea, ear infections, and/or rashes, which water quality test should be performed?

a. Phosphates
b. Total solids
c. Nitrates
d. Fecal coliform

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II. Scenario: Aragonite in Coral Reefs (30)

Answer all of the following questions. For calculation questions, show all work (including equations and substitutions) and round all answers to 2 decimal places.

1. Our oceans contain a natural buffer system using the carbonate and bicarbonate ion. Please write the equation for this buffer system below (5).
2. Describe what happens to the equilibrium of this system when excess carbon dioxide dissolves in the ocean as a result of burning fossil fuels. What happens to the pH of the ocean? How does the buffer compensate? (4)
3. The average ocean pH is about 8.1. Suppose a sample of ocean water has an H+ concentration of $8.0 \times 10^{-7} M$. How many times more acidic is this sample, compared to the average ocean pH? (4)
4. Calcifying organisms use dissolved carbonate and calcium to produce calcium carbonate through the following reaction: $Ca^{2+} + CO_3^{2-} \rightleftharpoons CaCO_3$. When excess fossil fuels are burned, what is the effect on $CaCO_3$ concentration? (2)
5. There are different forms of calcium carbonate, one notably being aragonite. What is aragonite? (2)
6. Therefore, given a Ca^{2+} concentration of $1.0 \times 10^{-4} M$, a lower-than-average CO_3^{2-} concentration of $1.0 \times 10^{-4} M$ and a solubility product of 1.4×10^{-8}, calculate the aragonite saturation state Ω_a. (3)
7. Is the sample of water from the previous question oversaturated or undersaturated with aragonite, and why? What does this mean for mineral formation? (3)
8. The saturation horizon is the depth at which water makes a transition from oversaturated to undersaturated. If surface waters contain more H^+ and less carbonate, as described in the ocean acidification case above, what happens to the saturation horizon and why? (2)
9. Knowing what you know now about the chemistry behind ocean acidification, what is the overall impact of ocean acidification on coral reefs? (5).

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III. Scenario: Dissolved Oxygen and BOD Testing

The lab that you are working in has obtained a sample of wastewater with bacteria in it, and asks you to perform a 5-day BOD (Biological Oxygen Demand) test.

First, you need to determine the dissolved oxygen (O_2) in the sample before and after the five-day incubation period, which can be done using the Winkler method. You titrate the sample in acidic solution with a standardized solution of sodium thiosulfate, $Na_2S_2O_3$. The following are the set of equations:

(1) $2Mn^{2+}(aq) + 4OH^-(aq) + O_2(aq) \rightarrow 2MnO_2(s) + 2H_2O(l)$
(2) $MnO_2(s) + 4H^+(aq) + 2I^-(aq) \rightarrow Mn^{2+}(aq) + I_2(aq) + 2H_2O(l)$
(3) $I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$

1. Combine the three equations above, and balance the overall equation (5).

Using this titration method, you titrated 10.00 mL samples of the water, one before and one after the five-day incubation period. This required 10.00 and 2.00 mL, respectively, of a 0.00100 M (mol/L) standard solution of $Na_2S_2O_3$.

2. Calculate the dissolved oxygen concentrations in the water samples before, and after incubation in mg/L of solution. Show all work. [$O_2 = 32.00\ g/mol$] (6)
3. Using your two calculated values of DO (before and after the 5-day period), calculate the BOD-5 (5-day biological oxygen demand) of diluted sample. The dilution was performed using 100 mL of the original sample in a 400 mL BOD bottle (4).

Another coworker in your lab took data for BOD of two different samples (sample A and sample B) for thirty days and generated a curve that looks something like the curve in Diagram G. Letters A and B represent the curve of samples A and B, respectively.

4. What is the name of the curve represented by the letter A? (1)
5. What is the name of the curve represented by the letter B? (1)
6. What is the difference between the two curves? (2)
7. What is the difference between the composition of sample A and sample B? (1)

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Finally, another coworker in your lab measured the DO and BOD of water in four water samples around a pollution zone of a sewage treatment plant. Diagram H shows an illustration of the four sites.

8. Predict what the DO and BOD curves looked like as distance from the pollution zone increases on your answer sheet. Plot DO and BOD as separate curves on the same graph and label them. Label the four zones (clean water, pollution zone, active decomposition zone, recovery zone). (10 pts)

Part IV: Salinometer Testing (10%)

1. What is the salinity, in percent mass/volume, of Sample A?
2. What is the salinity, in percent mass/volume, of Sample B?

yay congrats you're all done :)